

- 5 (a) transverse waves have vibrations that are perpendicular / normal
to the direction of energy travel B1
longitudinal waves have vibrations that are parallel
to the direction of energy travel B1 [2]
- (b) vibrations are in a single direction M1
either applies to transverse waves
or normal to direction of wave energy travel
or normal to direction of wave propagation A1 [2]
- (c) (i) 1. amplitude = 2.8 cm B1 [1]
2. phase difference = 135° or 0.75π rad or $\frac{3}{4}\pi$ rad or 2.36 radians
(three sf needed)
numerical value M1
unit A1 [2]
- (ii) amplitude = 3.96 cm (4.0 cm) A1 [1]
- 6 (a) (i) greater deflection M0